

Qualification Pack



Fiber Installation, Testing and Commissioning Technician

QP Code: TEL/Q4107

Version: 4.0

NSQF Level: 4

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TEL/Q4107: Fiber Installation, Testing and Commissioning Technician

Brief Job Description

A Fiber Installation, Testing and Commissioning Technician is responsible for overseeing on-site optical fiber installation, including cable preparation, splicing, connectorization, testing, and troubleshooting while ensuring adherence to industry standards and safety protocols. The individual oversees technicians and splicers, ensuring timely project completion and maintaining high-quality service (QoS) standards.

Personal Attributes

The individual should have strong analytical and problem-solving skills, attention to detail, and effective communication abilities to coordinate with team members, supervisors, and stakeholders.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [TEL/N4126: Fiber Construction, Performance and Selection Criteria](#)
2. [TEL/N4127: Fiber connectorisation, splicing and first level checks](#)
3. [TEL/N4128: Cable Installation Procedures and Practices](#)
4. [TEL/N4129: Preparing Cables for Termination and Splicing](#)
5. [TEL/N4130: Fiber Testing and Troubleshooting](#)
6. [TEL/N4131: Work Safety Practices with Fiber Optics](#)
7. [TEL/N9111: Follow sustainability practices in telecom cabling operations](#)
8. [DGT/VSQ/N0101: Employability Skills \(30 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Telecom
Sub-Sector	Passive Infrastructure
Occupation	Operations and Maintenance - Passive Infrastructure
Country	India

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NSQF Level	4
Credits	17
Aligned to NCO/ISCO/ISIC Code	NCO-2015/7422.0804
Minimum Educational Qualification & Experience	12th grade Pass OR Completed 2nd year of the 3-year diploma after 10 (in Fiber Optic Technology/ Electrical Er./ Networking and Communication Technology) OR 10th grade pass with 3 Years of experience in the operations and maintenance of passive infrastructure OR Previous relevant Qualification of NSQF Level (3.0) with 3 Years of experience in the optical fiber domain OR Previous relevant Qualification of NSQF Level (3.5) with 1.5 years of experience in the optical fiber domain
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	17 Years
Last Reviewed On	NA
Next Review Date	30/04/2028
NSQC Approval Date	08/05/2025
Version	4.0
Reference code on NQR	QG-04-TL-04086-2025-V2-TSSC
NQR Version	2

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TEL/N4126: Fiber Construction, Performance and Selection Criteria

Description

This unit is about understanding different fiber types, the specifications of passive components like Fiber Distribution Management System (FDMS), joint enclosures, patch cords, pigtailed, 0dB adapters, attenuators, and ensuring compliance with deployment parameters. It covers fiber construction, transmission checks, performance evaluation, and cable selection criteria, including bend-sensitive fibers, ribbon cables, and multi-core fibers for higher data transmission

Scope

The scope covers the following :

- Conduct optical fiber construction and transmission checks
- Evaluate fiber performance parameters
- Identify and select suitable fiber types and identifiers
- Follow appropriate cable selection criteria

Elements and Performance Criteria

Conduct optical fiber construction and transmission checks

To be competent, the user/individual on the job must be able to:

- PC1.** review and confirm the construction specifications of fiber cables, including core, cladding, buffer coating, and bend-sensitive fibers and ribbon cables, based on project requirements
- PC2.** guide team members to differentiate between cable components such as fibers, strength members, jackets, and multi-core fibers for higher data transmission.
- PC3.** ensure appropriate handling practices are followed while working with strengthening members, rip cords, armored fibers, and ribbon cables
- PC4.** verify transmission characteristics of different fiber types, including multimode, single-mode, and bend-sensitive fibers
- PC5.** monitor and ensure compliance with installation best practices to minimize losses and maintain performance, considering ease of installation (multi-fibers, microducting, etc.)

Evaluate fiber performance parameters

To be competent, the user/individual on the job must be able to:

- PC6.** review and guide the determination of performance indicators of an optical fiber such as attenuation, bandwidth, fiber size, and impact of multi-core fibers on data transmission
- PC7.** analyze performance by interpreting characteristic charts and parameter values
- PC8.** assess sources of attenuation, including scattering, absorption, bending radius, bending losses, and bend-sensitive fiber characteristics
- PC9.** guide differentiation between speed and bandwidth and their impact on fiber performance for network optimization, including multi-core fiber advantages
- PC10.** establish a correlation between attenuation and wavelength for effective network planning and guide team accordingly

Identify and select suitable fiber types and identifiers

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To be competent, the user/individual on the job must be able to:

- PC11.** guide classification of fiber types based on their construction like zip cord, distribution, loose tube, breakout cables, and ribbon cables
- PC12.** categorize fiber types based on deployment requirements, including armored, aerial, direct burial, underwater fibers, and bend-sensitive fibers
- PC13.** supervise selection and deployment of appropriate fiber types according to application and environmental factors, ensuring ease of installation
- PC14.** ensure proper identification and application of standard colour codes for fiber identification by the team
- PC15.** verify compatibility of selected fiber types with passive optical components for seamless integration and provide guidance where required

Follow appropriate cable selection criteria

To be competent, the user/individual on the job must be able to:

- PC16.** guide the selection of fiber cables based on essential parameters like pulling strength, water resistance, rodent protection, and ease of installation (multi-fiber, microducting, etc.)
- PC17.** ensure proper implementation of grounding and bonding techniques for armored cables by the installation team
- PC18.** implement grounding and bonding techniques for armored cables to enhance safety and durability
- PC19.** monitor and ensure compliance with environmental and industry standards while selecting cables for various deployment conditions

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the industry standards and best practices for optical fiber deployment, including multi-core and ribbon cables
- KU2.** the factors affecting fiber performance, including attenuation, dispersion, modal characteristics, and bend-sensitive fiber considerations
- KU3.** the characteristics of single-mode and multi-mode fibers, their applications, and limitations, as well as multi-core fiber capabilities
- KU4.** the specifications and functions of passive optical components such as connectors, splices, splitters, and enclosures
- KU5.** the importance of wavelength cut-off and its role in fiber selection
- KU6.** the electrical hazards and risks associated with fiber installation in high-voltage environments
- KU7.** the impact of bending radius, mechanical stress, and environmental conditions on fiber performance, particularly for bend-sensitive fibers
- KU8.** the principles of optical transmission, including reflection, dispersion, and loss mechanisms
- KU9.** the procedures for testing and verifying fiber continuity, insertion loss, and return loss
- KU10.** compliance requirements for health, safety, and environmental regulations in fiber network installation, considering ease of installation factors such as multi-fibers and microducting

Generic Skills (GS)

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User/individual on the job needs to know how to:

- GS1.** document and maintain records of fiber installation, test results, and maintenance activities
- GS2.** read and interpret technical specifications, performance charts, and industry guidelines
- GS3.** coordinate with colleagues, vendors, and stakeholders for seamless installation and troubleshooting
- GS4.** follow structured procedures and protocols for fiber handling, installation, and safety compliance
- GS5.** apply critical thinking and problem-solving skills to identify and resolve fiber performance issues
- GS6.** manage multiple installation and testing tasks while ensuring accuracy and adherence to industry standards
- GS7.** adapt to work with emerging fiber technologies and evolving installation methodologies
- GS8.** collaborate with multidisciplinary teams, including civil engineers, network planners, and system integrators
- GS9.** follow customer service protocols to address client inquiries and ensure service quality

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Conduct optical fiber construction and transmission checks</i>	10	15	-	6
PC1. review and confirm the construction specifications of fiber cables, including core, cladding, buffer coating, and bend-sensitive fibers and ribbon cables, based on project requirements	2	3	-	2
PC2. guide team members to differentiate between cable components such as fibers, strength members, jackets, and multi-core fibers for higher data transmission.	2	3	-	1
PC3. ensure appropriate handling practices are followed while working with strengthening members, rip cords, armored fibers, and ribbon cables	2	3	-	1
PC4. verify transmission characteristics of different fiber types, including multimode, single-mode, and bend-sensitive fibers	2	3	-	1
PC5. monitor and ensure compliance with installation best practices to minimize losses and maintain performance, considering ease of installation (multi-fibers, microducting, etc.)	2	3	-	1
<i>Evaluate fiber performance parameters</i>	8	15	-	5
PC6. review and guide the determination of performance indicators of an optical fiber such as attenuation, bandwidth, fiber size, and impact of multi-core fibers on data transmission	2	3	-	1
PC7. analyze performance by interpreting characteristic charts and parameter values	2	3	-	1
PC8. assess sources of attenuation, including scattering, absorption, bending radius, bending losses, and bend-sensitive fiber characteristics	1	3	-	1
PC9. guide differentiation between speed and bandwidth and their impact on fiber performance for network optimization, including multi-core fiber advantages	2	3	-	1

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. establish a correlation between attenuation and wavelength for effective network planning and guide team accordingly	1	3	-	1
<i>Identify and select suitable fiber types and identifiers</i>	8	15	-	5
PC11. guide classification of fiber types based on their construction like zip cord, distribution, loose tube, breakout cables, and ribbon cables	1	3	-	1
PC12. categorize fiber types based on deployment requirements, including armored, aerial, direct burial, underwater fibers, and bend-sensitive fibers	2	3	-	1
PC13. supervise selection and deployment of appropriate fiber types according to application and environmental factors, ensuring ease of installation	2	3	-	1
PC14. ensure proper identification and application of standard colour codes for fiber identification by the team	2	3	-	1
PC15. verify compatibility of selected fiber types with passive optical components for seamless integration and provide guidance where required	1	3	-	1
<i>Follow appropriate cable selection criteria</i>	4	5	-	4
PC16. guide the selection of fiber cables based on essential parameters like pulling strength, water resistance, rodent protection, and ease of installation (multi-fiber, microducting, etc.)	1	1	-	1
PC17. ensure proper implementation of grounding and bonding techniques for armored cables by the installation team	1	2	-	1
PC18. implement grounding and bonding techniques for armored cables to enhance safety and durability	1	1	-	1
PC19. monitor and ensure compliance with environmental and industry standards while selecting cables for various deployment conditions	1	1	-	1
NOS Total	30	50	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	TEL/N4126
NOS Name	Fiber Construction, Performance and Selection Criteria
Sector	Telecom
Sub-Sector	Passive Infrastructure
Occupation	Operations and Maintenance - Passive Infrastructure
NSQF Level	4
Credits	4
Version	5.0
Last Reviewed Date	08/05/2025
Next Review Date	30/04/2028
NSQC Clearance Date	08/05/2025

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TEL/N4127: Fiber connectorisation, splicing and first level checks

Description

This unit is about carrying out fiber end connections and splicing using various splicing techniques (mechanical, fusion, ribbon) and undertaking first level/immediate performance checks

Scope

The scope covers the following :

- Connector types and their use
- Mechanical splicing
- Fusion splicing
- Ribbon splicing
- First-level/immediate performance checks
- Record and report

Elements and Performance Criteria

Connector types and their use

To be competent, the user/individual on the job must be able to:

- PC1.** identify connectors based on color codes as per industry standards (TIA 568) and fiber type (single-mode, multimode)
- PC2.** review and guide the selection of connector types (ST, SC, FC/PC, MT, LC, MPO/MTP) based on application requirements and site conditions
- PC3.** verify and guide assessment of the effect of polish type (Flat, PC, UPC, APC) on insertion loss and return loss to ensure performance optimization
- PC4.** monitor and ensure proper execution of connector termination activities in field environments by the team, ensuring inspection and cleaning for optimal performance
- PC5.** identify and rectify common connector issues such as dirt contamination, misalignment, or end-face damage
- PC6.** incorporate pre-terminated solutions where applicable to eliminate the need for splicing
- PC7.** use Multi-Fiber Push-On (MPO) connectors in high-density cabling and data center environments

Mechanical splicing

To be competent, the user/individual on the job must be able to:

- PC8.** prepare fiber for splicing, including stripping the jacket, dressing buffer tubes and fibers, and removing buffer coatings
- PC9.** clean fibers using appropriate industry-standard solvents and tools
- PC10.** perform precise fiber cleaving using calibrated cleavers
- PC11.** execute mechanical splicing using elastomeric or other advanced splicing techniques
- PC12.** use pre-terminated solutions to reduce splicing needs and installation time

Fusion splicing

To be competent, the user/individual on the job must be able to:

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- PC13.** prepare fiber for fusion splicing, following the applicable steps
- PC14.** operate fusion splicing equipment, selecting the correct splicing program, performing arc calibration, and checking for dust contamination
- PC15.** conduct fusion splicing on various fiber types, ensuring minimal splice loss
- PC16.** review and validate splicing integrity using automated loss estimation tools and confirm compliance with performance standards
- PC17.** ensure the use of latest splicing machines and tools (e.g. heat-shrink sleeves) by the team for enhanced ease of splice and accuracy

Ribbon splicing

To be competent, the user/individual on the job must be able to:

- PC18.** prepare fibers for ribbon splicing using appropriate ribbon jacket strippers and alignment tools
- PC19.** cleave ribbon fibers using specialized ribbon cleavers
- PC20.** perform ribbon splicing using precision splicing equipment
- PC21.** validate and guide the assessment of splice integrity through loss estimation and inspection in ribbon splicing
- PC22.** apply ribbon fiber splicing in high-bandwidth networks

First-level/immediate performance checks

To be competent, the user/individual on the job must be able to:

- PC23.** perform immediate post-splicing tests using visual fault locator (VFL) and Optical Time Domain Reflectometer (OTDR)
- PC24.** monitor and guide evaluation of splices for defects such as white line, offset, diameter mismatch, bubbles, or bulging and ensure necessary corrective action is taken
- PC25.** identify root causes of improper splicing and take corrective actions
- PC26.** verify optical loss parameters measured by the team and ensure compliance with industry standards

Record and report

To be competent, the user/individual on the job must be able to:

- PC27.** follow standard data recording and reporting formats
- PC28.** document test results, performance parameters, cable drum markings, and splice location records
- PC29.** review and maintain accurate logs of splicing, losses, and corrective actions
- PC30.** validate and submit records and reports to appropriate authorities for validation and inspection

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the procedures related to fiber optic splicing, connectorisation, and testing
- KU2.** risks associated with non-compliance to standard procedures and quality assurance measures
- KU3.** escalation protocols for reporting technical issues or safety concerns

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- KU4.** types and specifications of fiber connectors, including emerging standards
- KU5.** attenuation factors, insertion loss, and return loss principles
- KU6.** optical loss minimization techniques and fiber handling best practices
- KU7.** operation and maintenance of stripping, cleaving, and splicing tools
- KU8.** common faults in fiber splicing and their troubleshooting techniques
- KU9.** industry standards for performance measurement and testing of fiber networks
- KU10.** occupational health, safety, and environmental guidelines applicable to fiber optic installations
- KU11.** the benefits of pre-terminated solutions (pre-attached connectors that don't require splicing) in reducing splicing efforts and installation time
- KU12.** the role of MPO connectors in modern fiber infrastructure and high-density networks
- KU13.** latest advancements in splicing machines and their impact on accuracy and efficiency
- KU14.** applications of ribbon fiber splicing for high-bandwidth and large-scale deployments

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and interpret technical documents such as fiber network diagrams, splice plans, and test reports
- GS2.** follow health and safety regulations and apply them in fiber splicing operations
- GS3.** use digital reporting tools and document work with accuracy
- GS4.** communicate technical details effectively with team members and supervisors
- GS5.** identify splicing faults and promptly implement corrective actions
- GS6.** collaborate with field technicians and network engineers to ensure smooth workflow
- GS7.** adapt to evolving fiber optic technologies and industry standards
- GS8.** troubleshoot unexpected challenges during fiber installations
- GS9.** manage time effectively to complete splicing and testing activities within project deadlines

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Connector types and their use</i>	7	10	-	5
PC1. identify connectors based on color codes as per industry standards (TIA 568) and fiber type (single-mode, multimode)	1	2	-	1
PC2. review and guide the selection of connector types (ST, SC, FC/PC, MT, LC, MPO/MTP) based on application requirements and site conditions	1	2	-	1
PC3. verify and guide assessment of the effect of polish type (Flat, PC, UPC, APC) on insertion loss and return loss to ensure performance optimization	1	2	-	1
PC4. monitor and ensure proper execution of connector termination activities in field environments by the team, ensuring inspection and cleaning for optimal performance	1	1	-	0.5
PC5. identify and rectify common connector issues such as dirt contamination, misalignment, or end-face damage	1	1	-	0.5
PC6. incorporate pre-terminated solutions where applicable to eliminate the need for splicing	1	1	-	0.5
PC7. use Multi-Fiber Push-On (MPO) connectors in high-density cabling and data center environments	1	1	-	0.5
<i>Mechanical splicing</i>	5	8	-	3
PC8. prepare fiber for splicing, including stripping the jacket, dressing buffer tubes and fibers, and removing buffer coatings	1	2	-	1
PC9. clean fibers using appropriate industry-standard solvents and tools	1	2	-	0.5
PC10. perform precise fiber cleaving using calibrated cleavers	1	2	-	0.5
PC11. execute mechanical splicing using elastomeric or other advanced splicing techniques	1	1	-	0.5

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. use pre-terminated solutions to reduce splicing needs and installation time	1	1	-	0.5
<i>Fusion splicing</i>	6	10	-	4
PC13. prepare fiber for fusion splicing, following the applicable steps	1	2	-	1
PC14. operate fusion splicing equipment, selecting the correct splicing program, performing arc calibration, and checking for dust contamination	1	2	-	1
PC15. conduct fusion splicing on various fiber types, ensuring minimal splice loss	1	2	-	0.5
PC16. review and validate splicing integrity using automated loss estimation tools and confirm compliance with performance standards	2	3	-	1
PC17. ensure the use of latest splicing machines and tools (e.g. heat-shrink sleeves) by the team for enhanced ease of splice and accuracy	1	1	-	0.5
<i>Ribbon splicing</i>	5	8	-	3
PC18. prepare fibers for ribbon splicing using appropriate ribbon jacket strippers and alignment tools	1	2	-	1
PC19. cleave ribbon fibers using specialized ribbon cleavers	1	2	-	0.5
PC20. perform ribbon splicing using precision splicing equipment	1	2	-	0.5
PC21. validate and guide the assessment of splice integrity through loss estimation and inspection in ribbon splicing	1	1	-	0.5
PC22. apply ribbon fiber splicing in high-bandwidth networks	1	1	-	0.5
<i>First-level/immediate performance checks</i>	4	8	-	3
PC23. perform immediate post-splicing tests using visual fault locator (VFL) and Optical Time Domain Reflectometer (OTDR)	1	2	-	1

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC24. monitor and guide evaluation of splices for defects such as white line, offset, diameter mismatch, bubbles, or bulging and ensure necessary corrective action is taken	1	2	-	1
PC25. identify root causes of improper splicing and take corrective actions	1	2	-	0.5
PC26. verify optical loss parameters measured by the team and ensure compliance with industry standards	1	2	-	0.5
<i>Record and report</i>	3	6	-	2
PC27. follow standard data recording and reporting formats	1	2	-	0.5
PC28. document test results, performance parameters, cable drum markings, and splice location records	1	2	-	0.5
PC29. review and maintain accurate logs of splicing, losses, and corrective actions	0.5	1	-	0.5
PC30. validate and submit records and reports to appropriate authorities for validation and inspection	0.5	1	-	0.5
NOS Total	30	50	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	TEL/N4127
NOS Name	Fiber connectorisation, splicing and first level checks
Sector	Telecom
Sub-Sector	Passive Infrastructure
Occupation	Operations and Maintenance - Passive Infrastructure
NSQF Level	4
Credits	3
Version	4.0
Last Reviewed Date	08/05/2025
Next Review Date	30/04/2028
NSQC Clearance Date	08/05/2025

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TEL/N4128: Cable Installation Procedures and Practices

Description

This unit is about installation procedures and practices for optical fiber cables

Scope

The scope covers the following :

- Pre-installation checks and processes
- Direct buried installation
- Underground (duct) installation
- Aerial installation

Elements and Performance Criteria

Pre-installation checks and processes

To be competent, the user/individual on the job must be able to:

- PC1.** plan and oversee pre-construction survey of the cable placing route, ensuring identification and mitigation of potential risks
- PC2.** inspect and test optical fiber cables using an OTDR to assess quality and detect any faults
- PC3.** perform pre-installation cable inspection to identify visible damage, compliance, and packaging integrity
- PC4.** verify key parameters such as pulling tension, bending radius, cable length, and splicing length at termination points to ensure readiness for installation
- PC5.** execute duct rodding, testing, and cleaning to ensure unobstructed cable installation, including the integration of micro ducting conduits for optimized capacity and scalability
- PC6.** ensure compliance with regulatory and safety requirements for the installation area, ensuring proper documentation for future maintenance

Direct buried installation

To be competent, the user/individual on the job must be able to:

- PC7.** guide selection of appropriate cable type (single jacket, dual jacket, armored) suitable for site conditions
- PC8.** perform armor bonding and grounding to enhance cable protection and longevity
- PC9.** handle and install cables while maintaining correct bending radius and tension levels
- PC10.** use cost-effective techniques like micro trenching to minimize excavation costs
- PC11.** monitor execution of trenching, compacting, and back-filling activities to optimize cable protection, reduce cost and installation time, and minimize wastage and damage
- PC12.** install underground warning tapes and markers to ensure cable identification and safety
- PC13.** ensure site restoration activities are carried out effectively post-installation to maintain infrastructure integrity and maintain proper documentation for future reference

Underground (duct) installation

To be competent, the user/individual on the job must be able to:

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- PC14.** perform duct route verification, testing, and clearance assessment before cable installation
- PC15.** use safe and efficient cable pulling techniques using appropriate tools and accessories such as pulling grips and breakaway swivels
- PC16.** execute cable reel positioning and controlled pulling to avoid excessive tension and kinking
- PC17.** implement 'figure-8' cable storing methods to prevent microbends and attenuation issues
- PC18.** conduct cable blowing operations using compressed air to facilitate long-distance installations, incorporating micro ducting conduits to improve efficiency
- PC19.** evaluate duct integration, fill ratio, and friction coefficients, and guide the team to optimize cable placement and lifespan

Aerial Installation

To be competent, the user/individual on the job must be able to:

- PC20.** guide selection and confirmation of aerial cables with appropriate strength members and weather resistance for specific installations
- PC21.** follow cable handling best practices, ensuring compliance with bending radius and tension guidelines
- PC22.** install messenger strands and secure cables using clamps, lashing, or self-supporting methods
- PC23.** deploy and install self-supporting aerial cables for stability and longevity
- PC24.** monitor and verify aerial cable sag and tensioning adjustments for proper alignment and load distribution
- PC25.** apply grounding and lightning protection measures to enhance cable durability and network reliability
- PC26.** ensure proper documentation of installation processes for future maintenance and troubleshooting

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the principles of project management, including time estimation, task prioritization, and resource allocation
- KU2.** the consequences of deviating from standard installation procedures, including performance issues and safety risks
- KU3.** the protocols for reporting incidents, technical issues, or emergencies such as system failures, fire hazards, or cable damages
- KU4.** regulatory and clearance requirements for underground, buried, and aerial installations
- KU5.** industry standards and best practices for safety, health, and environmental compliance
- KU6.** different types of optical fiber cables, including single-mode, multimode, armored, dielectric, and ribbon cables
- KU7.** suitability of optical fiber cable types for specific environmental and operational conditions
- KU8.** importance of maintaining proper cable handling techniques to prevent damage, including avoiding excessive tension and bend violations
- KU9.** characteristics and applications of key installation tools such as cable blowing machines, pulling grips, and tension monitoring equipment

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- KU10.** trenching, ducting, and aerial infrastructure requirements, including mechanical protection and environmental considerations, with a focus on micro trenching for minimizing excavation costs
- KU11.** the advantages of micro ducting conduits in cable installations, including scalability, ease of upgrades, and cost-efficiency
- KU12.** proper trenching and backfilling techniques to reduce installation time, lower costs, minimize material wastage, and prevent damage
- KU13.** the importance of documenting every installation step to facilitate future maintenance and ensure operational continuity

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** interpret technical drawings, cable routing plans, and site survey reports to plan installation activities
- GS2.** read and understand manufacturer manuals, safety instructions, and compliance documentation related to optical fiber installation
- GS3.** document pre-installation checks, test results, and post-installation observations accurately
- GS4.** communicate effectively with team members, supervisors, and third-party contractors to coordinate installation processes
- GS5.** provide clear and concise technical explanations when troubleshooting installation issues
- GS6.** adapt communication style based on stakeholders, including non-technical personnel and regulatory authorities
- GS7.** apply problem-solving skills to address unexpected site challenges such as obstructions or cable damage
- GS8.** use digital tools and software for cable testing, documentation, and network monitoring
- GS9.** prioritize tasks efficiently to meet project deadlines and minimize service disruptions
- GS10.** uphold professional and ethical standards while ensuring compliance with safety regulations and quality benchmarks

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Pre-installation checks and processes</i>	8	15	-	8
PC1. plan and oversee pre-construction survey of the cable placing route, ensuring identification and mitigation of potential risks	2	3	-	2
PC2. inspect and test optical fiber cables using an OTDR to assess quality and detect any faults	1	2	-	1
PC3. perform pre-installation cable inspection to identify visible damage, compliance, and packaging integrity	1	2	-	1
PC4. verify key parameters such as pulling tension, bending radius, cable length, and splicing length at termination points to ensure readiness for installation	1	2	-	1
PC5. execute duct rodding, testing, and cleaning to ensure unobstructed cable installation, including the integration of micro ducting conduits for optimized capacity and scalability	2	4	-	2
PC6. ensure compliance with regulatory and safety requirements for the installation area, ensuring proper documentation for future maintenance	1	2	-	1
<i>Direct buried installation</i>	8	10	-	4
PC7. guide selection of appropriate cable type (single jacket, dual jacket, armored) suitable for site conditions	2	2	-	1
PC8. perform armor bonding and grounding to enhance cable protection and longevity	1	2	-	0.5
PC9. handle and install cables while maintaining correct bending radius and tension levels	1	1	-	0.5
PC10. use cost-effective techniques like micro trenching to minimize excavation costs	1	1	-	0.5

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. monitor execution of trenching, compacting, and back-filling activities to optimize cable protection, reduce cost and installation time, and minimize wastage and damage	1	2	-	0.5
PC12. install underground warning tapes and markers to ensure cable identification and safety	1	1	-	0.5
PC13. ensure site restoration activities are carried out effectively post-installation to maintain infrastructure integrity and maintain proper documentation for future reference	1	1	-	0.5
<i>Underground (duct) installation</i>	6	13	-	4
PC14. perform duct route verification, testing, and clearance assessment before cable installation	1	3	-	1
PC15. use safe and efficient cable pulling techniques using appropriate tools and accessories such as pulling grips and breakaway swivels	1	2	-	1
PC16. execute cable reel positioning and controlled pulling to avoid excessive tension and kinking	1	2	-	0.5
PC17. implement 'figure-8' cable storing methods to prevent microbends and attenuation issues	1	2	-	0.5
PC18. conduct cable blowing operations using compressed air to facilitate long-distance installations, incorporating micro ducting conduits to improve efficiency	1	2	-	0.5
PC19. evaluate duct integration, fill ratio, and friction coefficients, and guide the team to optimize cable placement and lifespan	1	2	-	0.5
<i>Aerial Installation</i>	8	12	-	4
PC20. guide selection and confirmation of aerial cables with appropriate strength members and weather resistance for specific installations	1	1	-	0.5
PC21. follow cable handling best practices, ensuring compliance with bending radius and tension guidelines	1	1	-	0.5

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC22. install messenger strands and secure cables using clamps, lashing, or self-supporting methods	2	2	-	1
PC23. deploy and install self-supporting aerial cables for stability and longevity	1	2	-	0.5
PC24. monitor and verify aerial cable sag and tensioning adjustments for proper alignment and load distribution	1	2	-	0.5
PC25. apply grounding and lightning protection measures to enhance cable durability and network reliability	1	2	-	0.5
PC26. ensure proper documentation of installation processes for future maintenance and troubleshooting	1	2	-	0.5
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	TEL/N4128
NOS Name	Cable Installation Procedures and Practices
Sector	Telecom
Sub-Sector	Passive Infrastructure
Occupation	Operations and Maintenance - Passive Infrastructure
NSQF Level	4
Credits	3
Version	6.0
Last Reviewed Date	08/05/2025
Next Review Date	30/04/2028
NSQC Clearance Date	08/05/2025

Qualification Pack

TEL/N4129: Preparing Cables for Termination and Splicing

Description

This unit is about outside plant installation procedures and practices for optical fiber cables, ensuring adherence to industry standards and best practices

Scope

The scope covers the following :

- Fiber optic cable preparation
- Fiber optic cable handling
- Cable slack management

Elements and Performance Criteria

Fiber optic cable preparation

To be competent, the user/individual on the job must be able to:

- PC1.** remove the outer jacket using specially designed stripping tools and techniques while ensuring fiber integrity
- PC2.** utilize rip-cord efficiently for accessing the cable core
- PC3.** identify, dress, and secure the strength member for reliable termination
- PC4.** use kelling's grip for effective cable strain relief
- PC5.** cut armoured cable using industry-standard cutting tools while ensuring safety
- PC6.** check and verify cable specifications before preparation activities to ensure compliance with project requirements and prevent damage
- PC7.** apply right cleaving methods and quality cleaning techniques and materials for optimal performance
- PC8.** ensure proper wrapping, closing, and protection of cables by the team to maintain cable integrity during and after preparation

Fiber optic cable handling

To be competent, the user/individual on the job must be able to:

- PC9.** position cable drums correctly to prevent damage during installation
- PC10.** lift, shift, and handle cable drums using manual and mechanical methods such as fork-lifts
- PC11.** roll and unwind drums properly to prevent twisting and kinking
- PC12.** ensure unloading and storage of cable drums in accordance with manufacturer guidelines to maintain cable integrity
- PC13.** inspect cables before and after handling to identify potential defects or damages
- PC14.** ensure the team uses high-quality splicing tools to achieve precise and durable fiber connections

Cable slack management

To be competent, the user/individual on the job must be able to:

- PC15.** review and validate slack calculations based on project needs and industry best practices

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- PC16.** secure slack using slack brackets, loops, or storage enclosures for long-term stability
- PC17.** monitor and ensure that cable slack is managed appropriately to avoid interference with performance or future maintenance
- PC18.** ensure accurate documentation of slack management procedures by the team for future reference in maintenance and troubleshooting

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** organizational processes and procedures for cable handling, termination, and splicing
- KU2.** risk and impact of not following defined procedures and work instructions
- KU3.** industry standards and regulatory guidelines for fiber optic cable installations
- KU4.** safety, health, and environmental regulations related to cable installation and handling
- KU5.** proper usage and maintenance of tools and equipment used in fiber optic cable preparation
- KU6.** impact of incorrect cable laying practices on network performance and longevity
- KU7.** best practices for securing, storing, and transporting fiber optic cables
- KU8.** cable characteristics such as bend radius, tensile strength, and attenuation factors
- KU9.** importance of accurate documentation and record-keeping in cable installation
- KU10.** quality assurance procedures and testing requirements
- KU11.** effective use of specially designed stripping tools and techniques for fiber integrity
- KU12.** importance of right cleaving methods and quality cleaning techniques for reliable splicing
- KU13.** role of high-quality splicing tools in achieving precise fiber optic connections
- KU14.** significance of proper wrapping, closing, and protection of cables to ensure longevity and reliability

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** interpret technical documents such as work instructions, manuals, and specifications
- GS2.** read and understand safety guidelines and industry regulations
- GS3.** document procedures, issues, and solutions in a clear and concise manner
- GS4.** communicate effectively with team members, supervisors, and external stakeholders
- GS5.** coordinate with third-party vendors and suppliers for cable handling and installation
- GS6.** use digital tools and software for documentation, reporting, and troubleshooting
- GS7.** plan and organize work to ensure timely and efficient completion of tasks
- GS8.** identify and solve common problems related to cable preparation and handling
- GS9.** adhere to workplace safety protocols and promote a safe working environment
- GS10.** adapt to technological advancements in fiber optic cable installation and maintenance

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Fiber optic cable preparation</i>	12	20	-	9
PC1. remove the outer jacket using specially designed stripping tools and techniques while ensuring fiber integrity	2	3	-	2
PC2. utilize rip-cord efficiently for accessing the cable core	2	3	-	1
PC3. identify, dress, and secure the strength member for reliable termination	2	3	-	1
PC4. use kellum's grip for effective cable strain relief	2	3	-	1
PC5. cut armoured cable using industry-standard cutting tools while ensuring safety	1	2	-	1
PC6. check and verify cable specifications before preparation activities to ensure compliance with project requirements and prevent damage	1	2	-	1
PC7. apply right cleaving methods and quality cleaning techniques and materials for optimal performance	1	2	-	1
PC8. ensure proper wrapping, closing, and protection of cables by the team to maintain cable integrity during and after preparation	1	2	-	1
<i>Fiber optic cable handling</i>	10	20	-	6
PC9. position cable drums correctly to prevent damage during installation	2	4	-	1
PC10. lift, shift, and handle cable drums using manual and mechanical methods such as fork-lifts	2	4	-	1
PC11. roll and unwind drums properly to prevent twisting and kinking	2	4	-	1
PC12. ensure unloading and storage of cable drums in accordance with manufacturer guidelines to maintain cable integrity	2	4	-	1

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. inspect cables before and after handling to identify potential defects or damages	1	2	-	1
PC14. ensure the team uses high-quality splicing tools to achieve precise and durable fiber connections	1	2	-	1
<i>Cable slack management</i>	8	10	-	5
PC15. review and validate slack calculations based on project needs and industry best practices	2	3	-	2
PC16. secure slack using slack brackets, loops, or storage enclosures for long-term stability	2	3	-	1
PC17. monitor and ensure that cable slack is managed appropriately to avoid interference with performance or future maintenance	2	2	-	1
PC18. ensure accurate documentation of slack management procedures by the team for future reference in maintenance and troubleshooting	2	2	-	1
NOS Total	30	50	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	TEL/N4129
NOS Name	Preparing Cables for Termination and Splicing
Sector	Telecom
Sub-Sector	Passive Infrastructure
Occupation	Operations and Maintenance - Passive Infrastructure
NSQF Level	4
Credits	2
Version	4.0
Last Reviewed Date	08/05/2025
Next Review Date	30/04/2028
NSQC Clearance Date	08/05/2025

Qualification Pack

TEL/N4130: Fiber Testing and Troubleshooting

Description

This unit covers the testing and troubleshooting of fiber optic networks, including the use of industry-standard test equipment and documentation of results

Scope

The scope covers the following :

- Fiber test parameters
- Test equipment
- Fault localization and troubleshooting techniques
- Record and report

Elements and Performance Criteria

Fiber test parameters

To be competent, the user/individual on the job must be able to:

- PC1.** measure optical parameters such as optical power, attenuation, return loss, and reflection at fiber cables and connectors using the latest methods such as LSPM, OTDR, VFL, and Optical Fiber (OF) microscope
- PC2.** measure power loss for different fiber types using single-ended and double-ended loss measurement techniques, ensuring accuracy with advanced connectors
- PC3.** identify fiber continuity and polarity using fiber tracers and light sources while incorporating preventive maintenance practices
- PC4.** analyze optical signal loss patterns and detect potential macro/micro bends, splices, and connector faults to proactively prevent failures
- PC5.** interpret test results against industry standards and project specifications, ensuring data accuracy and reliability for remote monitoring systems

Test equipment

To be competent, the user/individual on the job must be able to:

- PC6.** operate OTDR to locate faults, measure fiber length, and assess splice and connector losses with a focus on preventive maintenance
- PC7.** use an Optical Power Meter (OPM) and light source to measure insertion loss and power levels accurately, utilizing advanced connectors for better efficiency
- PC8.** perform bidirectional OTDR testing to validate fiber loss values and reflectance characteristics while ensuring optimal network performance
- PC9.** use a Visual Fault Locator (VFL) to identify macro bends and fiber breaks in short-range fiber networks, applying latest fault detection techniques
- PC10.** inspect fiber end faces and connectors using a fiber inspection scope and clean them as per industry guidelines to prevent contamination
- PC11.** perform Optical Loss Test Set (OLTS) measurements for link certification and conformance testing, integrating remote monitoring where applicable

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- PC12.** ensure proper calibration and maintenance of testing instruments as per manufacturer's guidelines, for long-term reliability

Fault localization and troubleshooting techniques

To be competent, the user/individual on the job must be able to:

- PC13.** validate analysis of OTDR traces by the team to differentiate between fiber breaks, connector reflections, and splice losses, ensuring effective fault detection
- PC14.** use remote monitoring systems for quicker fault detection
- PC15.** use appropriate loss measurement methods such as LSPM and VFL to isolate the source of signal degradation
- PC16.** oversee the analysis of OTDR traces by the team to differentiate between fiber breaks, connector reflections, and splice losses, ensuring effective fault detection
- PC17.** verify the impact of external factors like temperature, stress, and bending on fiber performance, implementing preventive maintenance strategies
- PC18.** coordinate with installation teams and oversee rectification of fiber defects ensuring compliance with fiber optic standards and usage of advanced connectors

Record and report

To be competent, the user/individual on the job must be able to:

- PC19.** record and maintain test parameters such as power, attenuation, link loss, and event location as per standard formats, incorporating data for remote monitoring systems
- PC20.** ensure accurate documentation of fiber inspection parameters, including connector cleanliness, test results, and corrective actions taken by the team
- PC21.** generate and review test reports and ensure availability of documentation for audits, compliance checks, and future reference with an emphasis on secure digital storage
- PC22.** ensure adherence to organizational guidelines for secure storage and retrieval of test data, maintaining data integrity and availability for remote monitoring

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** industry standards and best practices for plant fiber testing and troubleshooting, including the latest methods such as LSPM, OTDR, VFL, and of microscope
- KU2.** the procedures for testing, reporting, and troubleshooting fiber optic networks with a focus on preventive maintenance
- KU3.** importance of following standard procedures and guidelines to ensure network reliability while integrating remote monitoring for proactive issue detection
- KU4.** occupational health and safety regulations related to fiber optics handling and testing, ensuring proper use of advanced connectors
- KU5.** optical power measurement principles, including signal attenuation, reflection, and return loss, with the inclusion of remote monitoring capabilities
- KU6.** interpretation of OTDR traces and event maps for fiber fault analysis to efficiently locate and rectify issues
- KU7.** measurement units such as dB, dBm, and their conversion for accurate testing, ensuring consistency across testing methodologies

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- KU8.** calibration procedures and maintenance practices for fiber testing equipment, incorporating best practices for longevity and efficiency
- KU9.** fiber handling best practices to minimize contamination, bending losses, and damage while using advanced connectors for better durability
- KU10.** types of fiber cables and their performance characteristics in different environmental conditions, ensuring adaptability in various network setups

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and interpret OTDR traces, test reports, and technical specifications accurately
- GS2.** communicate effectively with colleagues, supervisors, and external vendors for testing coordination
- GS3.** document and maintain accurate records of test results, inspections, and corrective actions
- GS4.** follow safety guidelines and use protective gear while handling fiber optic cables and test instruments
- GS5.** plan and prioritize tasks to complete testing and troubleshooting activities within project timelines
- GS6.** analyze test data and correlate findings with network performance issues
- GS7.** use digital tools and software applications for documentation and report generation
- GS8.** identify and apply problem-solving techniques to resolve fiber optic issues efficiently
- GS9.** adhere to industry standards and organizational policies in fiber testing and maintenance
- GS10.** work collaboratively with field teams to ensure seamless installation, testing, and commissioning of fiber networks

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Fiber test parameters</i>	8	15	-	5
PC1. measure optical parameters such as optical power, attenuation, return loss, and reflection at fiber cables and connectors using the latest methods such as LSPM, OTDR, VFL, and Optical Fiber (OF) microscope	2	3	-	1
PC2. measure power loss for different fiber types using single-ended and double-ended loss measurement techniques, ensuring accuracy with advanced connectors	2	3	-	1
PC3. identify fiber continuity and polarity using fiber tracers and light sources while incorporating preventive maintenance practices	2	3	-	1
PC4. analyze optical signal loss patterns and detect potential macro/micro bends, splices, and connector faults to proactively prevent failures	1	3	-	1
PC5. interpret test results against industry standards and project specifications, ensuring data accuracy and reliability for remote monitoring systems	1	3	-	1
<i>Test equipment</i>	10	20	-	6
PC6. operate OTDR to locate faults, measure fiber length, and assess splice and connector losses with a focus on preventive maintenance	2	3	-	1
PC7. use an Optical Power Meter (OPM) and light source to measure insertion loss and power levels accurately, utilizing advanced connectors for better efficiency	2	3	-	1
PC8. perform bidirectional OTDR testing to validate fiber loss values and reflectance characteristics while ensuring optimal network performance	2	3	-	1
PC9. use a Visual Fault Locator (VFL) to identify macro bends and fiber breaks in short-range fiber networks, applying latest fault detection techniques	1	3	-	0.5

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. inspect fiber end faces and connectors using a fiber inspection scope and clean them as per industry guidelines to prevent contamination	1	3	-	1
PC11. perform Optical Loss Test Set (OLTS) measurements for link certification and conformance testing, integrating remote monitoring where applicable	1	3	-	1
PC12. ensure proper calibration and maintenance of testing instruments as per manufacturer's guidelines, for long-term reliability	1	2	-	0.5
<i>Fault localization and troubleshooting techniques</i>	6	8	-	6
PC13. validate analysis of OTDR traces by the team to differentiate between fiber breaks, connector reflections, and splice losses, ensuring effective fault detection	1	1	-	1
PC14. use remote monitoring systems for quicker fault detection	1	1	-	1
PC15. use appropriate loss measurement methods such as LSPM and VFL to isolate the source of signal degradation	1	2	-	1
PC16. oversee the analysis of OTDR traces by the team to differentiate between fiber breaks, connector reflections, and splice losses, ensuring effective fault detection	1	1	-	1
PC17. verify the impact of external factors like temperature, stress, and bending on fiber performance, implementing preventive maintenance strategies	1	1	-	1
PC18. coordinate with installation teams and oversee rectification of fiber defects ensuring compliance with fiber optic standards and usage of advanced connectors	1	2	-	1
<i>Record and report</i>	6	7	-	3
PC19. record and maintain test parameters such as power, attenuation, link loss, and event location as per standard formats, incorporating data for remote monitoring systems	2	3	-	1

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC20. ensure accurate documentation of fiber inspection parameters, including connector cleanliness, test results, and corrective actions taken by the team	1	2	-	1
PC21. generate and review test reports and ensure availability of documentation for audits, compliance checks, and future reference with an emphasis on secure digital storage	1	1	-	0.5
PC22. ensure adherence to organizational guidelines for secure storage and retrieval of test data, maintaining data integrity and availability for remote monitoring	2	1	-	0.5
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	TEL/N4130
NOS Name	Fiber Testing and Troubleshooting
Sector	Telecom
Sub-Sector	Passive Infrastructure
Occupation	Operations and Maintenance - Passive Infrastructure
NSQF Level	4
Credits	2
Version	4.0
Last Reviewed Date	08/05/2025
Next Review Date	30/04/2028
NSQC Clearance Date	08/05/2025

Qualification Pack

TEL/N4131: Work Safety Practices with Fiber Optics

Description

This unit is about work safety practices whilst working with fiber optics, with specific updates to include safety protocols for Fiber-to-the-Home (FTTH/X) installations

Scope

The scope covers the following :

- Safety practices for working with fiber optics during FTTH/X installations
- Industry standards and guidelines

Elements and Performance Criteria

Safety practices for working with fiber optics during FTTH/X installations

To be competent, the user/individual on the job must be able to:

- PC1.** follow appropriate eye-safety measures whilst working with laser-emitting devices, including Optical Network Terminals (ONTs) and splicing equipment
- PC2.** safely handle bare fiber, broken ends, and scraps, ensuring proper disposal to prevent accidental injury
- PC3.** follow the recommended fire safety practices, especially when using high-voltage arc fusion splicers and other heating tools
- PC4.** adhere to electrical safety norms when working with fiber cables installed alongside electrical cables and active power sources
- PC5.** follow laser safety rules, including proper handling of class 1m and higher laser devices used in FTTH networks
- PC6.** use and maintain appropriate safety gear, such as gloves, boots, and protective eyewear, in accordance with worksite requirements
- PC7.** follow industry standards and guidelines (e.g., ANSI Z136.2 for laser safety, ISO standards for fiber optic installations)
- PC8.** maintain health and safety records accurately, including incident reports and compliance checklists
- PC9.** identify and mitigate hazards associated with installation sites, such as confined spaces, sharp edges, and high temperatures
- PC10.** examine and safely handle pre-terminated fiber assemblies and connectors to prevent contamination or damage
- PC11.** apply safe cable routing techniques to avoid damage to existing infrastructure, such as gas pipelines, electrical cables, and water pipelines

Industry standards and guidelines

To be competent, the user/individual on the job must be able to:

- PC12.** follow national and international guidelines for fiber optic installations, such as TIA/EIA-568, ITU-T standards, and BICSI guidelines

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- PC13.** comply with regulatory and environmental norms related to FTTH/X deployment, including e-waste management and proper waste disposal practices
- PC14.** identify and report non-compliance issues or deviations from industry standards to the supervisor

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** construction of fiber optics and methods for protecting fibers from environmental damage
- KU2.** the latest PPE required for fiber optic installations such as safety glasses, cut-resistant gloves, protective clothing, safety footwear, hard hats, hearing protection, high-visibility vests, and respiratory protection
- KU3.** the benefits of using PPE such as ensuring safety, preventing injuries, reducing workplace accidents, and ensuring compliance with occupational safety regulations
- KU4.** the safety features, limitations, and maintenance protocols for protective equipment and gear
- KU5.** laser safety guidelines, including risk levels of various laser classes used in fiber optics
- KU6.** the hazards associated with optical fibers, such as microshards and laser exposure, and safe disposal practices for fiber scraps and hazardous materials
- KU7.** layout of associated services in the work area (gas pipelines, electrical cables, sewage lines, water pipelines, etc.) to avoid consequential damage during FTTH installation
- KU8.** key industry standards and guidelines for fiber optic installations (e.g., ANSI/TIA, ITU-T, ISO)
- KU9.** compliance requirements for environmental regulations and e-waste management in FTTH installations
- KU10.** contact details of government agencies and regulatory bodies responsible for safety, health, and security
- KU11.** importance of maintaining accurate and timely documentation to ensure compliance with industry and organizational standards
- KU12.** procedures for conducting periodic audits to ensure adherence to safety standards
- KU13.** manufacturer-supplied MSDS for safe handling of fiber optic materials and chemicals
- KU14.** legislative requirements and organizational procedures for health, safety, and security, and the roles and responsibilities of individuals
- KU15.** different types of health and safety hazards in FTTH/X installation sites
- KU16.** limits of personal responsibility for dealing with hazards and escalating critical issues
- KU17.** different emergency situations, including handling accidental fiber cuts, exposure to high voltage, and adhering to emergency procedures
- KU18.** the importance of maintaining high standards of health, safety, and security
- KU19.** the implications of non-compliance with health, safety, and security protocols on individuals and organizations

Generic Skills (GS)

User/individual on the job needs to know how to:

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- GS1.** read and interpret necessary documents, including safety manuals, standards, and compliance guidelines
- GS2.** understand and follow operational health and safety instructions, memos, and reports
- GS3.** read and comprehend technical specifications related to FTTH/X installations
- GS4.** communicate effectively with colleagues, supervisors, and stakeholders to report hazards, seek guidance, and ensure smooth operations
- GS5.** liaise and coordinate with third-party vendors and contractors to ensure adherence to safety protocols
- GS6.** document safety incidents, near misses, and compliance checklists clearly and accurately
- GS7.** adapt to new technologies and safety equipment introduced in FTTH/X installations
- GS8.** manage time effectively to prioritize safety-related tasks without compromising installation timelines
- GS9.** identify and report unsafe conditions or practices to the appropriate authority
- GS10.** use digital tools and software for safety monitoring, documentation, and compliance tracking

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Safety practices for working with fiber optics during FTTH/X installations</i>	20	30	-	15
PC1. follow appropriate eye-safety measures whilst working with laser-emitting devices, including Optical Network Terminals (ONTs) and splicing equipment	2	3	-	2
PC2. safely handle bare fiber, broken ends, and scraps, ensuring proper disposal to prevent accidental injury	2	3	-	2
PC3. follow the recommended fire safety practices, especially when using high-voltage arc fusion splicers and other heating tools	2	3	-	1
PC4. adhere to electrical safety norms when working with fiber cables installed alongside electrical cables and active power sources	2	3	-	1
PC5. follow laser safety rules, including proper handling of class 1m and higher laser devices used in FTTH networks	2	3	-	2
PC6. use and maintain appropriate safety gear, such as gloves, boots, and protective eyewear, in accordance with worksite requirements	2	3	-	2
PC7. follow industry standards and guidelines (e.g., ANSI Z136.2 for laser safety, ISO standards for fiber optic installations)	2	3	-	1
PC8. maintain health and safety records accurately, including incident reports and compliance checklists	2	3	-	1
PC9. identify and mitigate hazards associated with installation sites, such as confined spaces, sharp edges, and high temperatures	2	2	-	1
PC10. examine and safely handle pre-terminated fiber assemblies and connectors to prevent contamination or damage	1	2	-	1

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. apply safe cable routing techniques to avoid damage to existing infrastructure, such as gas pipelines, electrical cables, and water pipelines	1	2	-	1
<i>Industry standards and guidelines</i>	10	20	-	5
PC12. follow national and international guidelines for fiber optic installations, such as TIA/EIA-568, ITU-T standards, and BICSI guidelines	4	8	-	2
PC13. comply with regulatory and environmental norms related to FTTH/X deployment, including e-waste management and proper waste disposal practices	3	6	-	2
PC14. identify and report non-compliance issues or deviations from industry standards to the supervisor	3	6	-	1
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	TEL/N4131
NOS Name	Work Safety Practices with Fiber Optics
Sector	Telecom
Sub-Sector	Passive Infrastructure
Occupation	Operations and Maintenance - Passive Infrastructure
NSQF Level	3
Credits	1
Version	5.0
Last Reviewed Date	08/05/2025
Next Review Date	30/04/2028
NSQC Clearance Date	08/05/2025

Qualification Pack

TEL/N9111: Follow sustainability practices in telecom cabling operations

Description

This unit covers the application of sustainable and environmentally responsible practices in telecom cabling operations. It includes identifying recyclable and hazardous materials, implementing effective waste management strategies, optimizing resource usage, and ensuring compliance with environmental regulations.

Scope

The scope covers the following :

- Identify recyclable, reusable, and hazardous materials
- Follow waste management, recycling, and disposal protocols
- Optimize material and energy usage in cabling work
- Comply with environmental and regulatory standards

Elements and Performance Criteria

Identify recyclable, reusable, and hazardous materials

To be competent, the user/individual on the job must be able to:

- PC1.** identify different types of materials used in cabling operations, such as optical fiber, connectors, protective ducts, and joint enclosures
- PC2.** segregate materials into recyclable (e.g., plastic ducts, fiber offcuts, packaging), reusable (e.g., spare connectors, splitters), and hazardous (e.g., lead-based cables, old batteries, chemical adhesives) categories
- PC3.** follow workplace procedures for storing recyclable materials separately from hazardous and general waste
- PC4.** label and store hazardous waste safely to prevent contamination or accidents during handling and disposal
- PC5.** ensure all recyclable and hazardous waste is recorded in logs for traceability and compliance

Follow waste management, recycling, and disposal protocols

To be competent, the user/individual on the job must be able to:

- PC6.** adhere to applicable SOPs for safe handling and disposal of non-recyclable and hazardous materials, such as fiber shards, cable sheaths, and chemical adhesives
- PC7.** coordinate safe disposal of waste materials at designated collection points or through approved vendors, as per organizational policies and environmental guidelines
- PC8.** reduce waste by minimizing excess material use and reusing components wherever possible
- PC9.** maintain clean and organized work sites to prevent environmental contamination and promote workplace safety
- PC10.** check for and report any issues related to improper waste disposal, unauthorized dumping, or environmental hazards to the relevant authority

Optimize material and energy usage in cabling work

To be competent, the user/individual on the job must be able to:

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- PC11.** plan cabling work to minimize material wastage by accurately measuring fiber optic lengths, ducts, and accessories before cutting or installing
- PC12.** use energy-efficient tools and machinery where possible, ensuring minimal power consumption without compromising work efficiency
- PC13.** ensure proper maintenance of cabling tools and equipment to reduce unnecessary repairs and replacements that contribute to waste
- PC14.** optimize the use of trenching and ducting by following efficient layout designs to minimize excavation and reduce material consumption
- PC15.** use and promote eco-friendly materials (such as low-impact protective coatings and biodegradable packaging) wherever feasible

Comply with environmental and regulatory standards

To be competent, the user/individual on the job must be able to:

- PC16.** follow national and local environmental laws, regulations, and workplace policies related to telecom cabling operations
- PC17.** maintain proper documentation of sustainability-related activities, including logs of disposed and recycled materials, to meet regulatory and audit requirements
- PC18.** conduct periodic self-audits to verify compliance with environmental policies
- PC19.** educate team members on best practices for sustainability, including waste segregation, material reuse, and responsible energy consumption
- PC20.** report any violations of environmental policies, hazardous material spills, or unsafe disposal practices to the designated supervisor or regulatory body

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** organizational policies on sustainability, waste reduction, and material reuse in telecom infrastructure projects
- KU2.** the procedures for recycling, hazardous waste handling, and safe disposal of telecom-related materials
- KU3.** the importance of sustainability in long-term infrastructure planning, including the impact of telecom waste on the environment
- KU4.** the classification of materials used in optical fiber cabling, including recyclable, reusable, and hazardous components
- KU5.** standard waste management procedures for telecom operations, including segregation, labeling, and disposal methods
- KU6.** the methods to reduce material wastage, such as accurate measurements, careful handling of fiber optic cables, and optimized trenching techniques
- KU7.** the environmental hazards associated with improper disposal of optical fibers, batteries, and chemical adhesives
- KU8.** the regulations and compliance requirements for hazardous material disposal under national and international environmental laws
- KU9.** energy-efficient work practices, including low-power tools, optimized route planning, and reduced excavation techniques

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- KU10.** the importance of proper record-keeping for disposal and recycling to ensure compliance and accountability

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate effectively sustainability practices and compliance requirements to team members and supervisors
- GS2.** identify and implement solutions for reducing material wastage and improving recycling efforts
- GS3.** select the best disposal or recycling methods based on regulatory compliance and workplace policies
- GS4.** determine the environmental impact of cabling operations and suggest improvements
- GS5.** work effectively with waste management teams, regulatory bodies, and supervisors to ensure compliance
- GS6.** accurately document materials used, disposed of, or recycled for audit and compliance purposes
- GS7.** prioritize sustainability initiatives while meeting project deadlines
- GS8.** incorporate new sustainable materials and methods into existing work processes

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Identify recyclable, reusable, and hazardous materials</i>	8	15	-	5
PC1. identify different types of materials used in cabling operations, such as optical fiber, connectors, protective ducts, and joint enclosures	2	3	-	1
PC2. segregate materials into recyclable (e.g., plastic ducts, fiber offcuts, packaging), reusable (e.g., spare connectors, splitters), and hazardous (e.g., lead-based cables, old batteries, chemical adhesives) categories	2	3	-	1
PC3. follow workplace procedures for storing recyclable materials separately from hazardous and general waste	2	3	-	1
PC4. label and store hazardous waste safely to prevent contamination or accidents during handling and disposal	1	3	-	1
PC5. ensure all recyclable and hazardous waste is recorded in logs for traceability and compliance	1	3	-	1
<i>Follow waste management, recycling, and disposal protocols</i>	6	15	-	5
PC6. adhere to applicable SOPs for safe handling and disposal of non-recyclable and hazardous materials, such as fiber shards, cable sheaths, and chemical adhesives	2	3	-	1
PC7. coordinate safe disposal of waste materials at designated collection points or through approved vendors, as per organizational policies and environmental guidelines	1	3	-	1
PC8. reduce waste by minimizing excess material use and reusing components wherever possible	1	3	-	1
PC9. maintain clean and organized work sites to prevent environmental contamination and promote workplace safety	1	3	-	1

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. check for and report any issues related to improper waste disposal, unauthorized dumping, or environmental hazards to the relevant authority	1	3	-	1
<i>Optimize material and energy usage in cabling work</i>	8	10	-	5
PC11. plan cabling work to minimize material wastage by accurately measuring fiber optic lengths, ducts, and accessories before cutting or installing	2	2	-	1
PC12. use energy-efficient tools and machinery where possible, ensuring minimal power consumption without compromising work efficiency	2	2	-	1
PC13. ensure proper maintenance of cabling tools and equipment to reduce unnecessary repairs and replacements that contribute to waste	2	2	-	1
PC14. optimize the use of trenching and ducting by following efficient layout designs to minimize excavation and reduce material consumption	1	2	-	1
PC15. use and promote eco-friendly materials (such as low-impact protective coatings and biodegradable packaging) wherever feasible	1	2	-	1
<i>Comply with environmental and regulatory standards</i>	8	10	-	5
PC16. follow national and local environmental laws, regulations, and workplace policies related to telecom cabling operations	2	2	-	1
PC17. maintain proper documentation of sustainability-related activities, including logs of disposed and recycled materials, to meet regulatory and audit requirements	2	2	-	1
PC18. conduct periodic self-audits to verify compliance with environmental policies	2	2	-	1
PC19. educate team members on best practices for sustainability, including waste segregation, material reuse, and responsible energy consumption	1	2	-	1

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC20. report any violations of environmental policies, hazardous material spills, or unsafe disposal practices to the designated supervisor or regulatory body	1	2	-	1
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	TEL/N9111
NOS Name	Follow sustainability practices in telecom cabling operations
Sector	Telecom
Sub-Sector	
Occupation	Generic
NSQF Level	4
Credits	1
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	30/04/2028
NSQC Clearance Date	08/05/2025

Qualification Pack

DGT/VSQ/N0101: Employability Skills (30 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

PC1. understand the significance of employability skills in meeting the job requirements

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.

Basic English Skills

To be competent, the user/individual on the job must be able to:

PC4. speak with others using some basic English phrases or sentences

Communication Skills

To be competent, the user/individual on the job must be able to:

PC5. follow good manners while communicating with others

PC6. work with others in a team

Qualification Pack

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

PC7. communicate and behave appropriately with all genders and PwD

PC8. report any issues related to sexual harassment

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

PC9. use various financial products and services safely and securely

PC10. calculate income, expenses, savings etc.

PC11. approach the concerned authorities for any exploitation as per legal rights and laws

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

PC12. operate digital devices and use its features and applications securely and safely

PC13. use internet and social media platforms securely and safely

Entrepreneurship

To be competent, the user/individual on the job must be able to:

PC14. identify and assess opportunities for potential business

PC15. identify sources for arranging money and associated financial and legal challenges

Customer Service

To be competent, the user/individual on the job must be able to:

PC16. identify different types of customers

PC17. identify customer needs and address them appropriately

PC18. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC19. create a basic biodata

PC20. search for suitable jobs and apply

PC21. identify and register apprenticeship opportunities as per requirement

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use basic spoken English language

KU6. Do and dont of effective communication

KU7. inclusivity and its importance

KU8. different types of disabilities and appropriate communication and behaviour towards PwD

KU9. different types of financial products and services

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- KU10.** how to compute income and expenses
- KU11.** importance of maintaining safety and security in financial transactions
- KU12.** different legal rights and laws
- KU13.** how to operate digital devices and applications safely and securely
- KU14.** ways to identify business opportunities
- KU15.** types of customers and their needs
- KU16.** how to apply for a job and prepare for an interview
- KU17.** apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate effectively using appropriate language
- GS2.** behave politely and appropriately with all
- GS3.** perform basic calculations
- GS4.** solve problems effectively
- GS5.** be careful and attentive at work
- GS6.** use time effectively
- GS7.** maintain hygiene and sanitisation to avoid infection

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. understand the significance of employability skills in meeting the job requirements	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	1	3	-	-
PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC4. speak with others using some basic English phrases or sentences	-	-	-	-
<i>Communication Skills</i>	1	1	-	-
PC5. follow good manners while communicating with others	-	-	-	-
PC6. work with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	1	-	-
PC7. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC8. report any issues related to sexual harassment	-	-	-	-
<i>Financial and Legal Literacy</i>	3	4	-	-
PC9. use various financial products and services safely and securely	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. calculate income, expenses, savings etc.	-	-	-	-
PC11. approach the concerned authorities for any exploitation as per legal rights and laws	-	-	-	-
<i>Essential Digital Skills</i>	4	6	-	-
PC12. operate digital devices and use its features and applications securely and safely	-	-	-	-
PC13. use internet and social media platforms securely and safely	-	-	-	-
<i>Entrepreneurship</i>	3	5	-	-
PC14. identify and assess opportunities for potential business	-	-	-	-
PC15. identify sources for arranging money and associated financial and legal challenges	-	-	-	-
<i>Customer Service</i>	2	2	-	-
PC16. identify different types of customers	-	-	-	-
PC17. identify customer needs and address them appropriately	-	-	-	-
PC18. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	1	3	-	-
PC19. create a basic biodata	-	-	-	-
PC20. search for suitable jobs and apply	-	-	-	-
PC21. identify and register apprenticeship opportunities as per requirement	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0101
NOS Name	Employability Skills (30 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	2
Credits	1
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Element/ Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each Element/ PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.
6. To pass the Qualification Pack assessment, every trainee should score the Recommended Pass % aggregate for the QP.
7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.

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Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
TEL/N4126.Fiber Construction, Performance and Selection Criteria	30	50	-	20	100	15
TEL/N4127.Fiber connectorisation, splicing and first level checks	30	50	-	20	100	15
TEL/N4128.Cable Installation Procedures and Practices	30	50	-	20	100	15
TEL/N4129.Preparing Cables for Termination and Splicing	30	50	-	20	100	15
TEL/N4130.Fiber Testing and Troubleshooting	30	50	-	20	100	10
TEL/N4131.Work Safety Practices with Fiber Optics	30	50	-	20	100	10
TEL/N9111.Follow sustainability practices in telecom cabling operations	30	50	0	20	100	10
DGT/VSQ/N0101.Employability Skills (30 Hours)	20	30	-	-	50	10
Total	230	380	0	140	750	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
NCVET	National Council for Vocational Education and Training
QP	Qualification Pack
MC	Model Curriculum
NSQF	National Skills Qualification Framework
NSQC	National Skills Qualification Committee
NOS	National Occupational Standards
NCO	National Classification of Occupations
ES	Employability Skills
TSSC	Telecom Sector Skill Council
TRAI	Telecom Regulatory Authority of India
ST	Straight Tip
SC	Subscriber Connector (also known as Square Connector or Standard Connector)
FC/PC	Ferrule Connector / Physical Contact
MT	Mechanical Transfer
LC	Lucent Connector (also known as Little Connector)
MPO/MTP	Multi-Fiber Push-On/ Multi-Fiber Termination Push-On
PC	Physical Contact
UPC	Ultra Physical Contact
APC	Angled Physical Contact
VFL	Visual Fault Locator
OTDR	Optical Time Domain Reflectometer

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LSPM	Light Source and Power Meter
OLTS	Optical Loss Test Set

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.